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| **PI Name:** | Ye, Jun Ph.D. | | |
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This is a grant for pre-flight ground-based research to explore emerging new techniques in coherent optical frequency synthesis and measurement, generation of precise atomic clock signals based on ultrahigh Q optical transitions, and distribution of stable optical clocks across communication networks, including free-space propagation. Our work envisions fundamental atomic clock experiments to be performed in space, and likely within the context of an actively-manned space station.
The proposed activities will have impacts to a number of important scientific and technological thrusts of NASA, including spaceborne optical interferometry, enhanced formation flying, and gravitational wave detection. Our main objective is to use ultra-stable lasers in space-based fundamental research, in particular for the gravitational and relativistic physics that has become more urgently relevant with LISA. Many general relativity experiments such as LISA and The Laser Interplanetary Ranging Experiment (LIRE) will need an unprecedented level of laser stability to

carry out successful missions. The proposed work should also make a strong impact to the cold atom research programs through the enhanced control of matter-field interaction.

We will study physical limitations of stable optical frequency sources such as diode-pumped solid state lasers stabilized by a combination of molecular and cavity resonances. We will employ our new understandings in the modulation process that are critical for signal recovery and the resultant system accuracy. Another concrete set of subtasks concerns active suppression of laboratory vibrations that affect all high-resolution measurements. Low-noise sensing, orthogonalization of the several vibrational degrees of freedom, and design of appropriate servo controllers are all part of this sub-task. Basically all measurements encounter environmental limitations as the measurement precision is enhanced. These thermal and acceleration-induced changes may occur naturally on longer timescales for experiments associated with a space-based research environment, particularly in a microgravity environment. We will also actively pursue a cold atom based optical frequency standard, building upon our recent advances in understandings of the Alkaline earth Sr atoms that offers a particularly attractive optical transition with potential precision and accuracy rivaling single-ion based systems. Once a laser system with an unprecedented stability is developed, we will transfer the stability of this optical oscillator to a wide range of desired optical and microwave spectra, using our recently developed methodology of establishing tight phase-coherent links among optical oscillators and between optical and microwave oscillators using femtosecond-laser-based optical comb technology. One concrete plan is to explore a portable fiber-laser-based system that will be self-referenced to produce stabilized optical and RF clock signals that can be used simultaneously for laser ranging and time keeping in spaceborne experiments. The significance of this development will also be reflected in the research area of space based laser cooling and atomic physics within NASA's microgravity fundamental physics program, for instance, in the production of ultracold atoms and in the study of coherent control and manipulation of atomic wavepackets. Finally, during the course of study we will pay close attentions to practical issues of transferring these ground-based technologies to adapt to the operating environment of spacecraft.

**Task Description:**

The most important milestones are: • Precision spectroscopy of super-narrow intercombination lines in ultra-cold Sr atoms. Explore high quality molecular transitions in cavity-enhanced configurations for portability. Build optical frequency standards with stability, precision and accuracy approaching and potentially surpassing the best current systems. • Accurate control of the modulation and signal recovery processes; Stability of cw optical local oscillator approaching 1 * 10-15 at 1 s. • The objectives for the work plan are very clear. We will explore several key systems for optical frequency standards including a super-narrow transition of ultra-cold alkaline earth atoms confined in optical lattices, strong and unperturbed transitions of iodine molecules in a cell, and narrow molecular overtone transitions in an enhancement cavity. We will perform fundamental research on ultrahigh resolution and supersensitive laser spectroscopy and acquire understandings in key issues facing frequency/phase stability of an optical oscillator. In a close connection to the first objective, we will enhance our understanding and improve our designs of cavity-based laser frequency stabilization, working on important issues such as thermal and vibration isolation of an optical interferometer, modulation purity and signal recovery, as well as the use of an optical cavity for direct stabilization of a femtosecond laser. Third, we will develop and utilize (and potentially deliver) a femtosecond laser system that is phase coherently stabilized to the cw laser with high frequency stability. In fact, we will transfer the unrivaled stability from one particular wavelength to all other colors that are desired. This system will be kept compact, power efficient and cost effective. As discussed above, this result should bring an immediate, strong impact to the mission of space interferometry, and a large array of NASA programs in general. The fourth objective of this study is to explore a portable fiber-laser-based system that will be self-referenced to produce stabilized optical and RF clock signals that can be used simultaneously for laser ranging and time keeping in space-borne experiments. The fifth objective is to explore the new form of coherent optical communication, where both the repetition frequency of the pulse train and the actual optical carrier itself will be used to make unambiguous and highly precise distance measurement. The significance of these developments will be reflected immediately in the research areas of space-based laser cooling and atomic physics within NASA's microgravity fundamental physics program. The question of how to achieve these goals is the main focus of detailed discussions presented in the next section. We note in advance that the technologies and understandings developed in our previous work funded by NASA will form some of the main corner stones that we build upon to make essential improvement and progresses. Our successful activities that are particularly worth mentioning at this stage include the work of phase stabilization of femtosecond lasers, demonstration of an optical frequency synthesizer and an optical atomic clock, tight synchronization and coherent pulse synthesis from separate ultrafast lasers, coherent phase transfer among separate femtosecond lasers, frequency stabilization of solid-state lasers to molecular transitions of iodine and acetylene, active cancellation of mechanical vibration on optical cavities, enhanced understanding of modulation purification for optical frequency stabilization, and finally an improved system design for an ultimate optical frequency standard based on cold Sr atoms. At last, we also wish to note that during the course of study we will pay close attentions to practical issues of transferring these ground-based technologies to adapt to the operating environment of spacecraft. Space interferometry is the unifying theme of many programs under study or development in NASA. To be able to collect light from scientific targets with a large aperture and then precisely combine different light beams interferometrically, it is necessary to have precision control of optical paths. Various levels of precision are needed, for example, Space Interferometry Mission (SIM) needs nanometer level control and stabilization of optical element positions to measure the relative positions and distances of stars: 1 nm is required to carry out successful experiments of star light nulling, 200 pm for wide-angle astrometry, and 50 pm for narrow-angle astrometry. Space Technology 3 (ST3) and Terrestrial Planet Finder (TPF) are missions to be carried out with separated spacecraft interferometry, leading to an elevated requirement on control and stability. Laser frequency stability on the order of 1 x 10-16 at 1 s will make these tasks realistic. The need for a more stable atomic clock will always be there as we try to the put the entire universe on a finer and finer map. Laser metrology between satellites will further improve the precision of enhanced formation flying [1]. In the program of Gravity Recovery and Climate Experiment (GRACE), the Earth's gravity field is being mapped by measuring the along track separation between two spacecraft. A follow-on program, EX-5, calls for a drag-free spacecraft and distance tracking of 100 nm. This enhanced capability will improve the measurement sensitivity to changes in the Earth's gravity field by an order of magnitude compared with GRACE.

The nation's fundamental physics research program can certainly benefit a great deal from the improved optical local oscillators. Many general relativity experiments such as LISA and The Laser Interplanetary Ranging Experiment (LIRE) will need an unprecedented level of laser stability to carry out successful missions. Such achievement can be made possible only through the continued high level of research on laser frequency stabilization. The proposed work should also make a strong impact to the cold atom research programs through the enhanced control of matter-field interaction.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

The research activities carried out in this proposal are ground based and have stimulated many exciting new directions in the fields of precision measurement, metrology, ultracold matters, and ultrafast technology. While the primary goal of this research is to provide a key technological basis for space-borne interferometry, the technical advances achieved under this grant will have far-reaching consequences to a number of earth based scientific endeavors, including ultrasensitive trace gas detection, the next generation atomic clock, precision length metrology, and coherent XUV generations etc.

We have made major progress in the proposed research areas for the past fiscal year of 2005.
On the ultrastable laser development, we have demonstrated state-of-the-art laser stabilization via high finesse optical cavities limited by fundamental mechanical thermal noise-related length fluctuations. This work gives a detailed empirical verification of the theoretical model on thermal noise of cavity mirrors, by comparing the frequency noise of lasers tightly locked to the resonances of a variety of rigid Fabry-Perot cavities of differing lengths and mirror substrate materials. The understandings we have gained from this work on the fundamental limits of Fabry-Perot reference cavities will impact planning and construction of next generation, ultrastable optical cavities. At the same time we have continued our exploration of the remarkable capability we have developed in connecting optical and radio frequency domains in a phase-coherent manner. Femtosecond lasers have been stabilized to produce exceptional performances in both short-term (in terms of individual optical comb linewidth) and long-term (in terms of optical frequency reproducibility and accuracy) time scales. In our recent work, we are able to transfer the optical coherence of a Hz-linewidth cw laser to every comb component of an octave-spanning femtosecond optical frequency comb. This capability has allowed us to directly compare the optical linewidth of two cw lasers, one operated at 1.06 micron, the other at 698 nm, demonstrating an effective optical heterodyne beat signal with a linewidth of 3 Hz across this vast optical spectral bandwidth. The same femtosecond optical frequency comb serves as the clockwork for the neutral atom based optical atomic clock in our laboratory.

To enable systematic evaluations of stability and accuracy of the optical atomic clock, we use the remotely transferred hydrogen maser signal from the National Institute of Standards and Technology, where the maser signal is evaluated against the primary frequency standard of a Cs fountain clock. We have demonstrated precise transfer of timing signals over a fiber link that is vital for both precision frequency metrology and highly stable timing synchronization. With fast, active cancellation of the fiber transmission noise, we have synchronized two mode-locked femtosecond fiber lasers, connected remotely via a 7 km fiber network, to an RMS timing jitter of 19 fs, observed over the entire Nyquist bandwidth (half of the repetition frequency of 93 MHz). This result is achieved in two steps. First, active cancellation of the fiber-transmission noise reduces timing jitter caused by path length fluctuations to a record level of 16 fs. Second, using a wide bandwidth actuator, the remote laser is synchronized to the incoming stable pulse train from the reference laser at 10 fs, a record for fiber lasers. These results are confirmed in an optical cross-correlation measurement performed independent of the feedback loop operated in the microwave domain. These results also demonstrate that we now have the capability to control the noise level of a mode-locked fiber laser to be at nearly the same level as a mode-locked Ti:Sapphire laser. The mode-locked fiber lasers will be the workhorse for the future system of a portable femtosecond laser system for simultaneous time-keeping (optical atomic clock) and distance ranging (see the next paragraph) tasks.

On related femtosecond frequency comb work, we have accomplished several important exploratory studies on the interactions between a femtosecond-laser based optical frequency comb and a high finesse, low dispersion, passive optical cavity. First, we have achieved direct stabilization of a frequency comb to a high-finesse passive optical cavity. Such performance represents the state-of-art in frequency/phase stabilization of a femtosecond laser and fully confirms the expectation that a highly stable, passive optical cavity can directly stabilize the repetition frequency and carrier-envelope phase of ultrashort pulses to a level that rivals that achieved by cw lasers. The second exciting development is that we have demonstrated a general technique for enhancement of femtosecond pulses via coherent buildup inside a high-finesse cavity. Starting with an actively stabilized, but otherwise standard mode-locked laser system, over 10 microJoule pulses are demonstrated with an enhancement factor exceeding 500. This is accomplished over a large optical bandwidth such that ultrafast pulses (50 fs) can be coherently enhanced. Thirdly, we have developed a precision measurement protocol to characterize mirror loss and dispersion properties. This technical capability is absolutely needed to improve our capability in producing large bandwidth, low loss and low dispersion mirrors.

The ability to efficiently couple and store mode-locked pulse trains to high finesse cavities opens avenues for many novel experiments that can take advantage of the broad spectrum and high peak-pulse intensities stored in the “femto-cavity”. There is a range of applications that can benefit from simple, cost effective systems capable of producing pulse energies in the microJoule to milliJoule range. Many applications would also benefit from higher repetition rates for enhanced signal acquisition capabilities. Our approach represents a flexible and general technique for enhancement of femtosecond pulse energies by nearly three orders of magnitude, applicable to ultrashort pulse trains in any spectral region where appropriate mirrors can be produced. This approach can be used to compliment recent high-energy femtosecond oscillators to potentially produce pulse energies >10 microJoule. Besides this more conventional aspect of pulse amplification, the femtosecond enhancement cavity allows a new, exciting opportunity to explore nonlinear optical interactions in an intracavity environment at the laser's original high repetition rate.

For the extreme nonlinear optics work performed inside a femtosecond enhancement cavity, we have explored the technical frontier in both the shortest possible EUV wavelength and the highest possible EUV power that can be obtained from our approach. Due to the small conversion efficiency of this process, a femtosecond enhancement cavity is ideally suited for high harmonic generation (HHG) as the driving pulse is continually “recycled” after each pass through the gas target. The coherent frequency comb produced in the EUV spectral region resulting from the intracavity generation of high-harmonics was coupled out using a ~700 μm-thick sapphire plate placed at Brewster's angle (for the IR) inside the cavity. The Fresnel reflection coefficient of the intracavity plate rises at shorter wavelengths, with a reflectivity of ~5-10% between 40 and 100 nm. However, this plate is often damaged when the level of HHG intensity is raised. This has prevented us from scaling this system up to higher powers, needed to generate higher harmonics and make this a practical source of EUV light. We have designed novel enhancement cavity configurations which will allow us to use more powerful lasers and efficiently generate harmonics much further into the EUV (well below 50 nm). The concept is technically very challenging, involving careful engineering of the mirror coatings that make up the cavity, custom apertures in the optical substrates, and good servo electronics.

**Task Progress:**

We have also pursued another important research direction, namely cavity-based wide-bandwidth sensitive spectroscopy. The spectroscopy is accomplished by efficiently coupling a broadband frequency comb into a high finesse optical cavity, creating simultaneous ring-down decays of hundreds of thousands of individual cavity modes, which are used to probe atomic and molecular resonances located inside the passive cavity. Spectroscopic systems capable of making the next generation of atomic and molecular measurements will require: A large spectral bandwidth allowing for the observation of global energy level structure of many different atomic and molecular species, high spectral resolution for the identification and quantitative analysis of individual spectral features, high sensitivity for detection of trace amounts of an atom or molecule and for recovery of weak spectral features such as dipole forbidden transitions, and finally, fast spectral acquisition time, which takes advantage of high sensitivity, for the observation of spectral changes due to changing environmental conditions, leading to the study of dynamics. Our approach addresses all of the mentioned system characteristics, yielding a powerful combination of bandwidth, sensitivity, resolution, and acquisition speed unmatched by any existing experimental techniques.

We have also made major advances towards our goal of realizing an ultimate optical atomic clock using super-narrow intercombination lines in ultra-cold Sr atoms. Ultracold Sr work has entered the exciting stage of ultraprecision spectroscopy after achieving a record low temperature (250 nK, half a single photon recoil temperature) and loading both fermionic and bosonic isotopes into 1D optical lattice tuned to the magic wavelength to cancel AC Stark shift. We have carried out the most precise spectroscopy of lattice optical clock to date, achieving a statistical error of 2.8 Hz.

Alkaline earth atoms loaded in a so-called magic wavelength optical clock now have the prospect of becoming an ultimate clock system, combining the merits of ultrahigh stability from a large ensemble of ultracold neutral atoms and high accuracy that has been traditionally enjoyed by single trapped ion systems. We have carried out a systematic study of this important system. Taking advantage of the available reference signal from the NIST Cs fountain clock serving as the primary time standard, we have achieved a statistical uncertainty of the absolute optical frequency measurement of 2.8 Hz, more than a factor of three better than the previous result. We have also performed a detailed analysis of systematic shifts (19 Hz uncertainty) in the absolute frequency measurement of 429 228 004 229 869 Hz. The high resolution permits an investigation of the optical lattice motional sideband structure. The local oscillator for this optical atomic clock is a stable diode laser with its Hz-level linewidth characterized across the optical spectrum using a femtosecond frequency comb.

Furthermore, we anticipate that the capability of performing ultrahigh resolution spectroscopy of ultracold atoms localized inside an optical lattice will open the door for a fascinating array of studies one can perform on the properties of atomic interactions and control. To provide first such example, we apply our spectroscopic probe to study the motional dynamics of Sr atoms in a magic wavelength optical lattice. We have initiated photoassociation experiments on ultracold Sr atoms loaded in an optical lattice. This will pave the way for understanding atom-atom interactions which will be important both for metrology work and for fundamental atomic physics.

SUPPLEMENTAL PROGRESS FOR FINAL REPORTING PERIOD

During the final period of this project spanning from January to September, 2006, we achieved more breakthrough results relevant to the proposed areas of research.

On the ultrastable laser development, we have demonstrated phase and frequency stabilization of a diode laser at the thermal noise limit of a passive optical cavity. The system is compact and exploits a cavity design that reduces vibration sensitivity. The sub-Hz laser is characterized by comparison to a second independent system with similar fractional frequency stability (1 x 10-15 at 1 s). Highly frequency-stabilized lasers are essential in high resolution spectroscopy and quantum measurements, optical atomic clocks, and quantum information science. To achieve superior stability with high bandwidth control, the laser output is traditionally servo locked to a highly isolated passive optical cavity using the Pound-Drever-Hall (PDH) technique. Typical limits to the resulting laser stability include vibration-induced cavity length fluctuation, photon detection shot noise, and thermal-mechanical noise of the passive cavity components. By thoughtful system design, the relative impact of these noise contributions can be adjusted. We employ a cavity-laser system that achieves high stability with an appropriate compromise between acceleration sensitivity and thermal noise. The compact, cavity-stabilized diode laser has a sub-Hz linewidth (at several seconds) and is thermal-noise-limited to a fractional instability of ~1 x 10-15 at time scales of 0.5 to 300 s. As further evidence of the laser's optical stability, we use it to interrogate an ultranarrow optical atomic transition in neutral atomic strontium. We resolve an optical transition linewidth of ~2 Hz, the narrowest optical atomic resonance observed to date.

The laser source in this work is a diode laser in an external cavity (ECDL) in the Littman configuration operating at 698 nm. The laser is first pre-stabilized to a simple optical cavity with finesse of ~10,000. The PDH stabilization is accomplished via feedback to the laser diode current and the laser cavity piezo transducer (PZT). The servo bandwidth is 2–3 MHz. The prestabilized laser light is first-order diffracted by an acousto-optic modulator (AOM) and fiber-coupled to a platform on which an ultrastable cavity resides. This platform is mounted on a commercially available passive vibration isolation unit. Both the platform and the isolation unit are within a temperature-controlled enclosure lined with acoustic-damping foam. On the platform, the prestabilized laser light is phase modulated by an electro-optic modulator (EOM) operating at 5 MHz. Approximately 10 μW of optical power is incident on the ultrastable cavity for PDH locking. Stabilization to this cavity is accomplished via feedback to the AOM and a PZT controlling the prestabilization cavity length. The servo bandwidth for this final locking stage is ~100 kHz. The useful output of the laser is delivered from a fiber port located near the ultrastable cavity on the same platform. The entire optical set up occupies less than 1 m3.

Furthermore, femtosecond lasers have been stabilized to produce exceptional performances in both short-term (in terms of individual optical comb linewidth) and long-term (in terms of optical frequency reproducibility and accuracy) time scales. In our recent work, we are able to transfer the optical coherence of a Hz-linewidth cw laser to every comb component of an octave-spanning femtosecond optical frequency comb. This capability has allowed us to directly compare the optical linewidth of two cw lasers, one operated at 1.06 micron, the other at 698 nm, demonstrating an effective optical heterodyne beat signal with a linewidth of 3 Hz across this vast optical spectral bandwidth. The same femtosecond optical frequency comb serves as the clockwork for the neutral atom based optical atomic clock in our laboratory. In recent months of 2006, we have improved the heterodyne beat linewidth to 200 mHz, a world record for the coherence of the optical frequency comb.

Ultracold Sr atoms confined in an optical lattice have now served as one of the most stable and accurate atomic clocks. Precision measurement with ultracold neutral atoms confined in a trapping potential (a magic wavelength optical lattice)

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| | shows superior optical coherence without any deleterious effects from motional degrees of freedom, revealing optical resonance linewidths at the hertz level with excellent signal to noise ratio. The resonance quality factor of 2.4 x 10(14) is the highest ever recovered in any form of coherent spectroscopy. The spectral resolution permits direct observation of the breaking of nuclear spin degeneracy for the 1S0 and 3P0 optical clock states of 87Sr under a small magnetic bias field. This optical NMR-like approach allows a precision measurement of the differential Landé g-factor between the two states, leading to an accurate determination of the field-free hyperfine interaction-induced state mixing in the 3P0 state and its metastable lifetime. The unprecedented optical atomic coherence demonstrated in our lab for the collective excitation of a large number of atoms will have a strong impact on quantum measurement and precision frequency metrology. Furthermore, the overall systematic uncertainty of the 1S0 - 3P0 clock transition has been characterized below 9E-16, representing the first time a neutral atom based optical clock has reached the same level of accuracy as the primary standard of fountain clock, with the fractional stability of the former far surpassing that of the latter. Work is progressing rapidly in this area. |
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